Beyond the Algorithm: Visualizing the Environmental Cost of AI

Team 1 | Advanced Data Visualization
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Challenge Overview

The Data Challenge

Every time someone uses an AI tool, there's a physical cost most people don't think about. Running a large language model requires electricity, and cooling the servers that power it requires water. At small scale that's not a big deal, but AI has scaled up fast. This dashboard challenge asks teams to visualize the environmental footprint of AI and data centers across three dimensions: energy consumption, water usage, and carbon emissions. The goal is to make something usually abstract and technical feel concrete and understandable.

The central question driving this challenge: how big is AI's environmental impact, and how does it compare to things we already have a sense of scale for?

Purpose and Intent

This is primarily an informational and analytical dashboard. The data should tell a story, but users should also be able to explore it on their own and form their own conclusions. A few questions that shape the intent:

- Why does this matter? Because data centers are being built right now at a record pace, and the public rarely gets a clear picture of what that means environmentally.
- Is there a claim worth investigating? Some argue AI companies are overstating their sustainability commitments while underreporting actual consumption. The data can help test that.
- Are there trends to show? Yes. Energy use from data centers has been climbing steadily, and projections suggest the growth is accelerating as AI workloads increase.
- Do we want one interpretation or many? Multiple. The data is genuinely uncertain in places, and the dashboard should reflect that rather than hide it.

Stakeholders

The most realistic audience for a class dashboard is curious students and instructors who want to engage with a current policy debate backed by real data. Beyond that, this type of visualization would be useful for environmental policy researchers, journalists covering tech and climate, and people inside tech companies who work on sustainability reporting. Anyone trying to evaluate whether AI companies' green pledges hold up against their actual resource use would find value in it.

Targeted Questions

Teams working on this challenge should try to address the following through their visualizations:

- How has data center energy consumption changed over time in the U.S. and globally?
- How does AI's electricity use compare to other energy-intensive industries like steel, aluminum, and aviation?
- How much water do data centers use for cooling, and how does that vary by region?
- What are the estimated carbon emissions tied to AI workloads, and how do they compare to other familiar sources?
- How much transparency do major tech companies actually provide in their sustainability reports, and do the numbers add up?

Project Plan

Overview

The plan below lays out four weeks of work. The focus is on getting one or two solid visualizations working well rather than building something overly complex. Based on the datasets available, most time will likely go toward cleaning and reshaping the EIA and Our World in Data files so they can be plotted cleanly in R. Starting with a line chart of energy use over time is the safest first step since the data for that is already in decent shape.

Week	Phase	Tasks	Deliverable
Week 1	Design	Download EIA CSV and Our World in Data energy files. Look through the Kaggle data center dataset to see what columns are actually usable. Sketch two or three dashboard layout ideas. Assign roles. Do a first pass at loading data into R and checking for problems.	Sketches + data loaded in R
Week 2	Prototype	Clean and reshape data: standardize units (convert everything to TWh), handle missing years, join datasets where needed. Build a first draft line chart of U.S. and global data center energy use over time. Make a rough summary table.	Working rough visualization
Week 3	Build	Refine the line chart. Add a second visual comparing AI energy or water use to other industries. Add explanatory text and source notes. Start thinking about how the dashboard layout reads as a whole.	Near-final dashboard draft
Week 4	Deploy	Share with a few classmates for feedback. Revise based on what they find confusing. Polish labels, color, and theme. Host on shinyapps.io. Final check on all citations and data source notes.	Live hosted dashboard

Team Roles

These are rough guidelines. Overlap is expected and fine, especially during the build week.

- Data Wrangler: downloads and cleans the datasets. Main job is getting the EIA and Our World in Data files into a format R can actually work with -- standardizing units, handling gaps, joining tables. Also responsible for documenting what was changed and why.
- Visualization Lead: owns the ggplot code. Handles the main chart designs, picks colors and themes, and makes sure the visuals are actually readable by someone who didn't build them.
- Writer/Analyst: writes the contextual text that sits alongside the visuals. Explains what the data shows, flags limitations, and handles citation formatting for the references page.
- Deployment Lead: gets the dashboard live on shinyapps.io or a similar platform. Coordinates the feedback round in week four and manages final revisions.

Dashboard Sketch

The layout below shows what we're imagining for the dashboard. The idea is three panels total: a trend line up top, an industry comparison in the middle, and a water/carbon breakdown at the bottom. Each panel answers a different question, but they all connect to the same central argument about AI's environmental footprint.

Beyond the Algorithm: The Environmental Cost of AI

Tracking energy, water, and carbon across U.S. and global data centers

PANEL 1: Line Chart -- Energy Use Over Time
U.S. and Global Data Center Electricity Consumption (2015-2030)
x: Year | y: TWh | two lines: U.S. vs Global | shaded band shows projection range post-2024
Sources: EIA Commercial Buildings Energy Consumption Survey, Our World in Data (IEA)

PANEL 2: Bar Chart -- Industry Comparison
Data Centers vs. Steel, Aluminum, Aviation, Chemicals (TWh, 2024)
horizontal bars per industry | data centers highlighted in accent color | label shows % of total U.S. electricity
Sources: IEA Energy and AI Report, EIA

PANEL 3A: Water Use	**PANEL 3B: Carbon Emissions**
Data center water consumption (billions of liters) vs. bottled water consumed globally	Estimated CO2 from AI workloads (megatons) compared to NYC annual emissions
Source: EESI, De Vries-Gao (2025)	Source: GAO-25-107172, De Vries-Gao (2025)

Data sources listed below each panel | Last updated: May 2026

The sketch is intentionally simple. Panels 1 and 2 answer the core energy questions. Panels 3A and 3B bring in water and carbon, which broaden the environmental angle and give the dashboard more depth without requiring completely different datasets. All four should be buildable with ggplot without needing anything too exotic.

Background

When people talk about the environmental impact of AI, the conversation usually stays pretty vague. Energy use gets mentioned, maybe carbon emissions, sometimes water. But the actual numbers rarely make it into the discussion in a way that feels grounded. That's a problem, because the decisions being made right now about where to build data centers, how to power them, and whether to require companies to disclose their consumption are all consequential and all depend on having a clearer picture of what's actually going on.

The energy side of this is the most documented. Data centers in the U.S. consumed around 176 to 183 terawatt-hours (TWh) of electricity in 2023, which is somewhere around 4% of total U.S. electricity demand. Globally, the figure was roughly 415 TWh that year, or about 1.5% of world electricity use. The International Energy Agency projects that could more than double to around 945 TWh globally by 2030, with AI being the main driver of that growth. For context, 415 TWh is roughly equivalent to the annual electricity consumption of a mid-sized country. The U.S. EIA tracks commercial building energy consumption through its Commercial Buildings Energy Consumption Survey (CBECS), which includes data centers as a category and is freely available for download. Our World in Data has also cleaned and reformatted the IEA's data center figures into a straightforward CSV, which is probably the easiest starting point for getting something into R quickly.

The water side is less covered but arguably just as important. Data centers use water in two main ways: directly for cooling systems (often called water usage effectiveness, or WUE), and indirectly through the water consumed in generating the electricity they run on. The Environmental and Energy Study Institute has reported that data centers' direct water use alone is substantial, and a 2025 study by researcher Alex de Vries-Gao found that AI systems could be responsible for between 312.5 and 764.6 billion liters of water use per year. That's roughly the same order of magnitude as all bottled water consumed globally in a single year, which is a striking comparison. Major tech companies largely do not disclose AI-specific water figures, and some have explicitly said they exclude indirect water use from their sustainability reports.

The carbon picture is also murky, partly because it depends heavily on what energy sources power the grid in different regions. The U.S. Government Accountability Office released a technology assessment in April 2025 (GAO-25-107172) specifically looking at generative AI's environmental and human effects. It found that estimates of AI's environmental impact are highly variable, in part because companies don't disclose enough data to verify them. The GAO report includes figures for estimated carbon emissions and energy equivalents for training large AI models, which are useful anchor points even if they're estimates. De Vries-Gao's 2025 study puts AI's potential annual carbon footprint at between 32.6 and 79.7 million tonnes of CO_2, roughly comparable to the annual emissions of a city the size of New York.

One tension worth acknowledging: some researchers and industry representatives argue that comparisons like these overstate the problem, because AI hardware has gotten dramatically more efficient per task even as total consumption has grown. There's something to that. But efficiency gains don't necessarily translate to lower total consumption when the scale of adoption is growing faster than the efficiency improvements. That's exactly the kind of claim a well-built visualization could help users think through.

One honest limitation to flag upfront: the Dryad dataset listed in our initial source list (doi:10.5061/dryad.73n5tb363) turns out to be a single-building energy management dataset from a Honda research facility in Europe, and probably isn't the right fit for a macro-level dashboard about AI and data centers. The primary datasets we plan to use are the EIA commercial energy data, the Our World in Data/IEA files, and the Kaggle data center dataset (pending verification of its columns). The GAO report, HBR article, EESI piece, and World Economic Forum article will be used as background citations and for specific figures that appear as annotations in the visualizations.

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